Year 8 Science

Mixtures and Compounds

Aim: To compare a mixture and a compound.
Materials: Magnet, iron filings, sulfur, iron sulfide, watch glasses, hand lens, teaspoons.
Method:

1. Place one teaspoon of the iron filings onto a watch glass, and one teaspoon of sulfur onto the other watch glass.
2. Test the iron and the sulfur for magnetism (by moving the magnet underneath the watch glass). **Do not put the magnet into the iron filings**.
3. Record how the iron and the sulfur are affected by the magnet.
4. Carefully smell the iron and then the sulfur. Record your observations.
5. Examine each substance using the hand lens. Record your observations.
6. Pour the iron filings onto the watch glass containing the sulfur. Gently mix them together.
7. Repeat steps 2 to 5 with the mixture.
8. Place a teaspoon of iron sulfide on the empty watch glass and repeat steps 2 to 5 with the iron sulfide.

Results:
Set up a table like the one below to record your observations.

	Iron filings	**Sulfur**	**Mixture**	**Compound – Iron sulfide**
Magnetism				
Smell				
Hand lens observations				

Questions:

1. Have any of the properties of the iron and sulfur changed by mixing them together?
2. Describe how you could separate the two substances.
3. Does the iron sulfide show any of the properties of the iron or the sulfur?
4. Could you separate the iron from the sulfur in the iron sulfide using the methods that you would use to separate the iron and the sulfur in the mixture?
5. What are the differences between the properties of the mixture and the compound?

6. Write the chemical formulas for: iron, sulfur, iron sulfide.

